

Programme	B. Sc. Computer Science				
Course Title	Social Network Analysis				
Type of Course	Elective				
Semester	VIII				
Academic Level	400 - 499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	Knowledge in Fundamentals of Data Mining				
Course Summary	The syllabus is prepared with the view of preparing the BSc Computer Science Graduates to build a basic understanding of what social network analysis is and how it can be applied. Topics covered include network structure and methods for social network analysis, link analysis and network community detection, information propagation on the web and some applications				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the basic notation and terminology used in social network analysis.	U	C	Instructor-created exams / Quiz
CO2	Compare and interpret social network structure, size and its connectivity pattern.	U	C	Instructor-created exams / Quiz
CO3	Discover community structure in complex network using statistical techniques	U	P	Instructor-created exams / Quiz
CO4	Apply link prediction techniques to discover new links in the social network	U	C	Instructor-created exams / Case studies
CO5	Describe influence in social media, perform recommendations	U	P	Instructor-created exams / Quiz Case studies
CO6	Perform Social Influence Analysis	U	P	Instructor-created exams / Quiz / Case studies
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C)# - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I	Introduction to Social Network Data Analytics		12	15
	1	Introduction to Social Network Analysis	2	
	2	Online social networks Research Issues and Topics	2	
	3	Statistical properties of social networks: Preliminaries	2	
	4	Static properties, Dynamic properties	3	
	5	Challenges of Social Network Streams	3	
II	Random Walks in Social Networks		12	15
	6	Random walks on Graphics, Walks based on proximity measures	2	
	7	Other graph based proximity measures	1	
	8	Graph theoretic measures for semi supervised learning	3	
	9	Clustering with random walk based measures	3	
	10	Applications in computer vision Text Analysis, Evaluation and datasets	1	
	11	Link prediction and data sources	2	
III	Community Discovery in Social Networks		12	20
	12	Communities in Context	2	
	13	Core Methods – KL Algorithm, Special algorithms	2	
	14	Markov Clustering, other approaches	2	
	15	Emerging Fields and problems : Community Discovery in dynamic networks	2	
	16	Heterogeneous networks, Directed networks,	2	
	17	Coupling content and relationship information for community discovery	2	
IV	Link Prediction in Social Networks		12	20
	18	Background, Feature based Link Prediction, Bayesian Probabilistic Models	3	
	19	Probabilistic Relational Models	2	
	20	Linear Algebraic Methods	2	
	21	Link Predictions: The Katz Score, Hitting & Commute Time	2	
	22	Rooted PageRank, SimRank	3	
V	Open Ended Module		12	
	CASE STUDY: Social Influence Analysis Influence Related Statistics, Social Similarity and Influence, Influence Maximization in Viral Marketing,			

Reference Books:

1. Charu.C. Aggarwal, Social Network Data Analytics, Springer Science+Business Media, LLC 2011.

2. R. Zafarani, M. A. Abbasi, and H. Liu, Social Media Mining: An Introduction, Cambridge University Press, 2014.
3. Krishna Raj P M, Ankith Mohan, K G Srinivasa ,Practical Social Network Analysis with Python , Springer Liu, Bing. Web data mining. Springer-Verlag Berlin Heidelberg, 2007.
4. Chakrabarti, Soumen. Mining the Web: Discovering knowledge from hypertext data. Morgan Kaufmann, 2003.
5. Scime, Anthony, ed. Web mining: applications and techniques. IGI Global, 2005

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO1	-	2	-	1	1	3						
CO2	-	3	-	1	1	2						
CO3	1	3	-	1	1	1						
CO4	1	2	-	-	1	1						
CO5	-	1	1	-	1	1						
CO6	-	1	1	-	1	1						

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓			✓
CO 2	✓			✓
CO 3	✓	✓		✓
CO 4		✓		✓
CO 5		✓		✓
CO 6			✓	